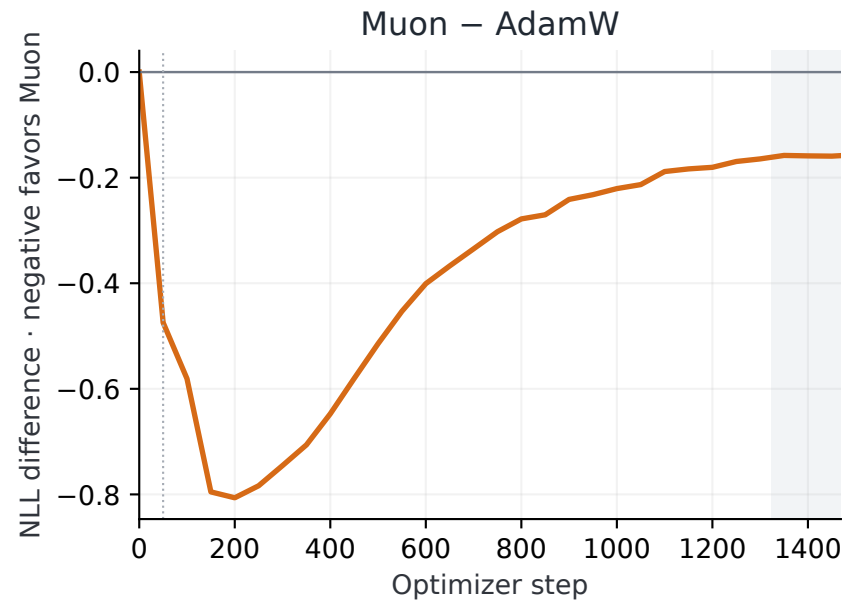
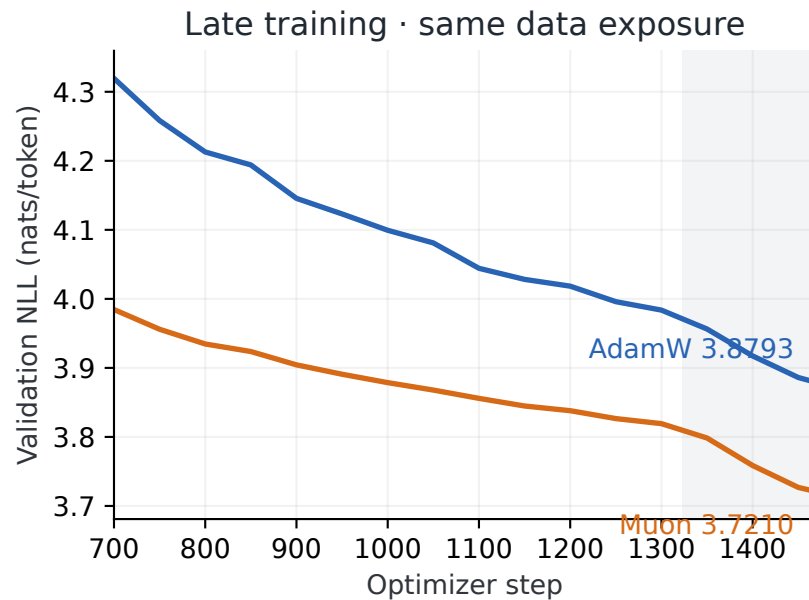
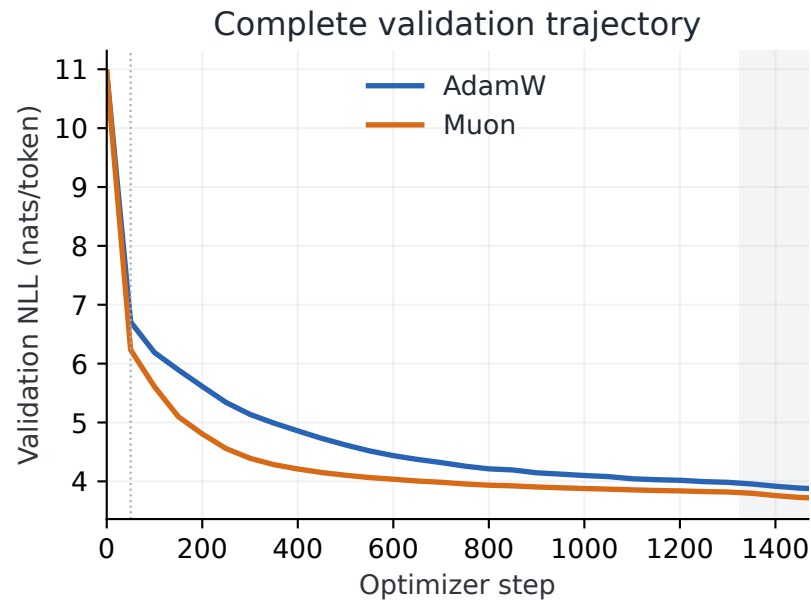


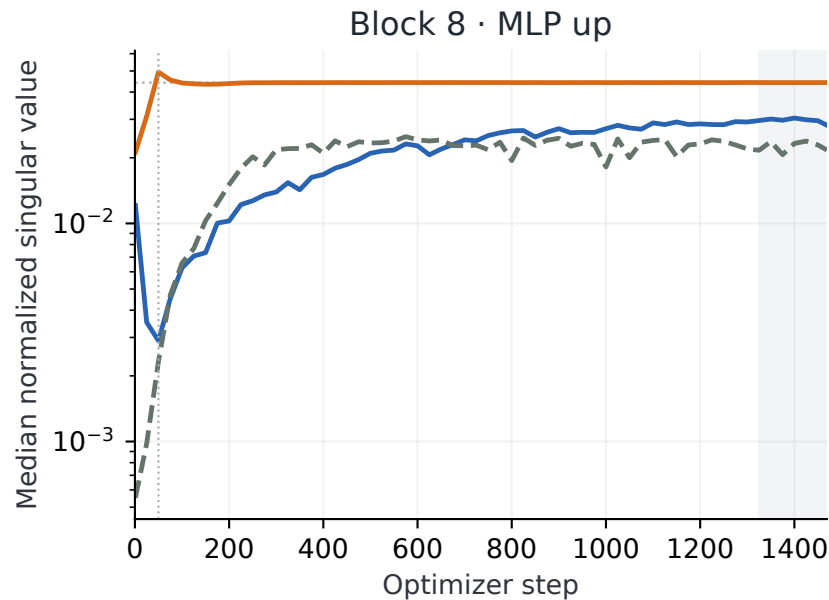
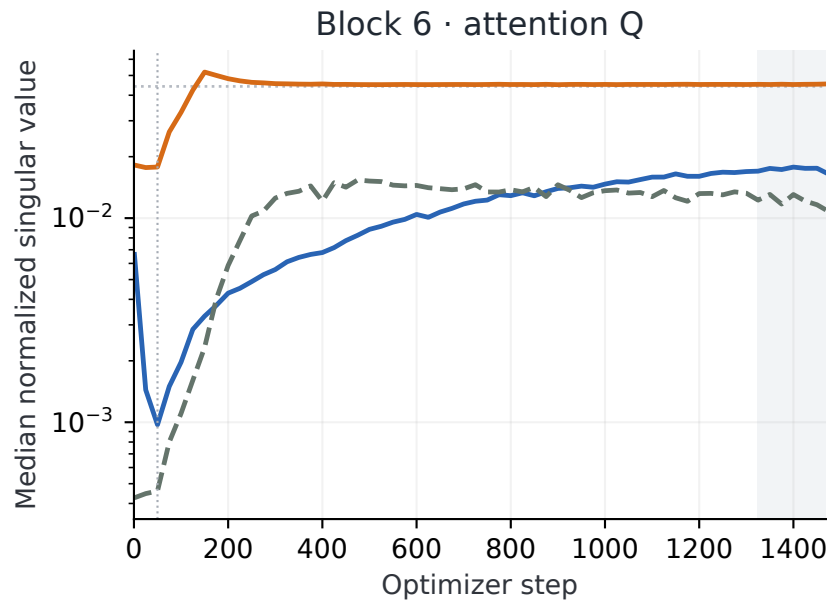
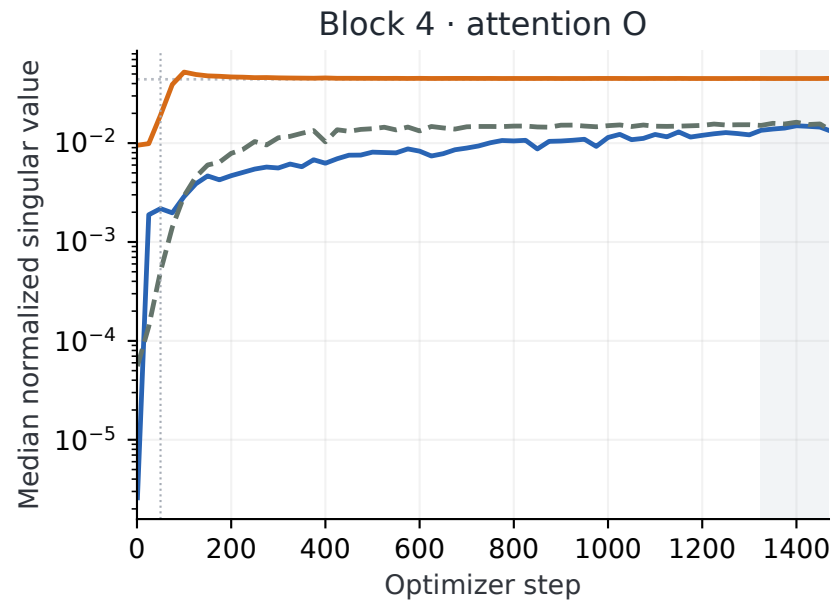
8 layers · Muon and AdamW validation loss



Same initialization, width 512, global batch and 1.540B tokens. One seed. Shading marks final-10% cooldown.
Recipes differ: AdamW LR 0.0012 throughout; Muon body LR 0.01 and auxiliary AdamW LR 0.002.

8 layers · spectral evolution at three fixed locations

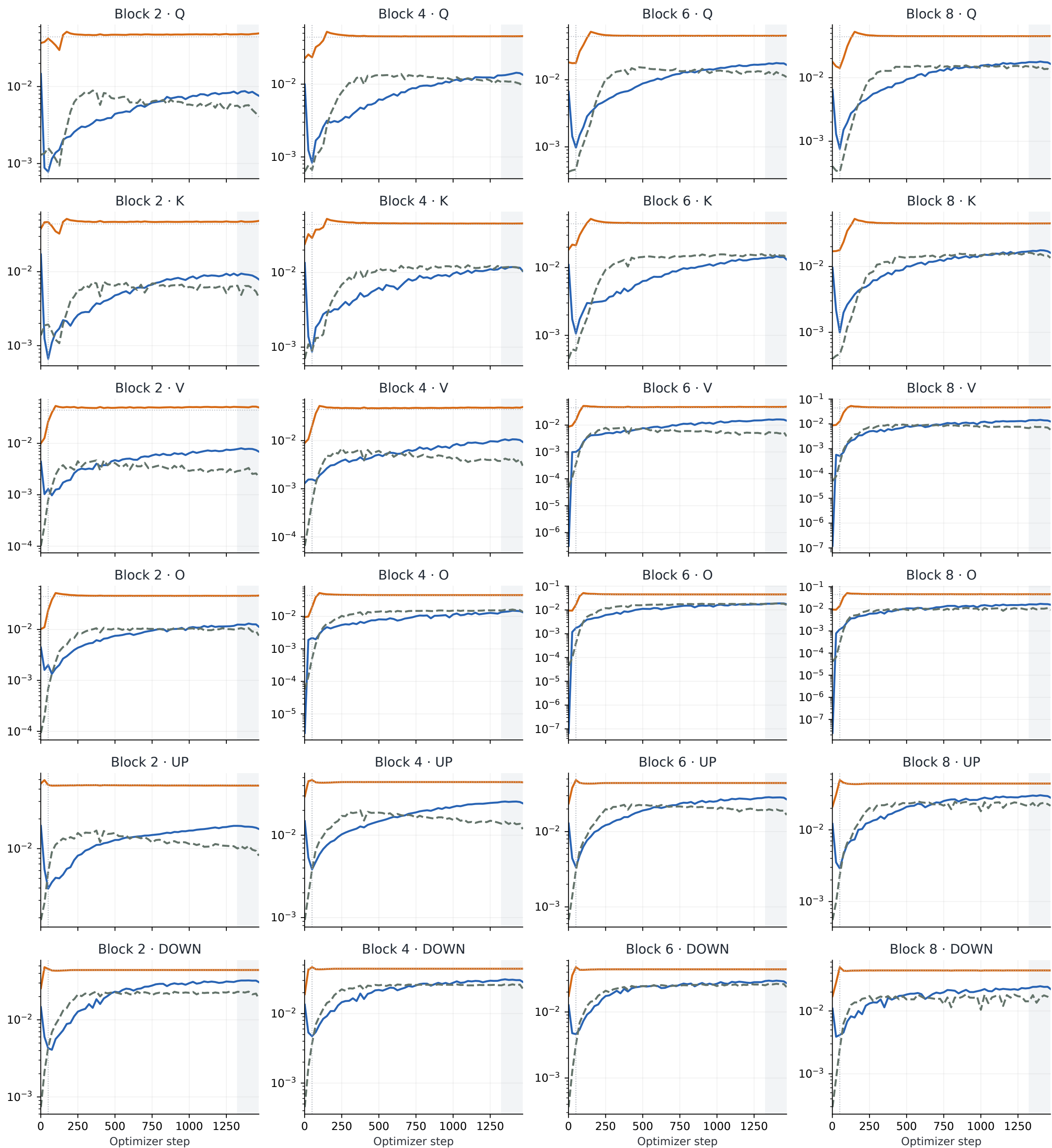
AdamW adaptive update Muon post-NS update Muon pre-NS momentum



Each matrix is divided by its own Frobenius norm. Dashed green is momentum, before NS; solid curves are updates. Dotted horizontal line: equal singular values, $1/\sqrt{512} \approx 0.0442$. LR and weight decay are excluded.

Median normalized singular value · all 24 matrices

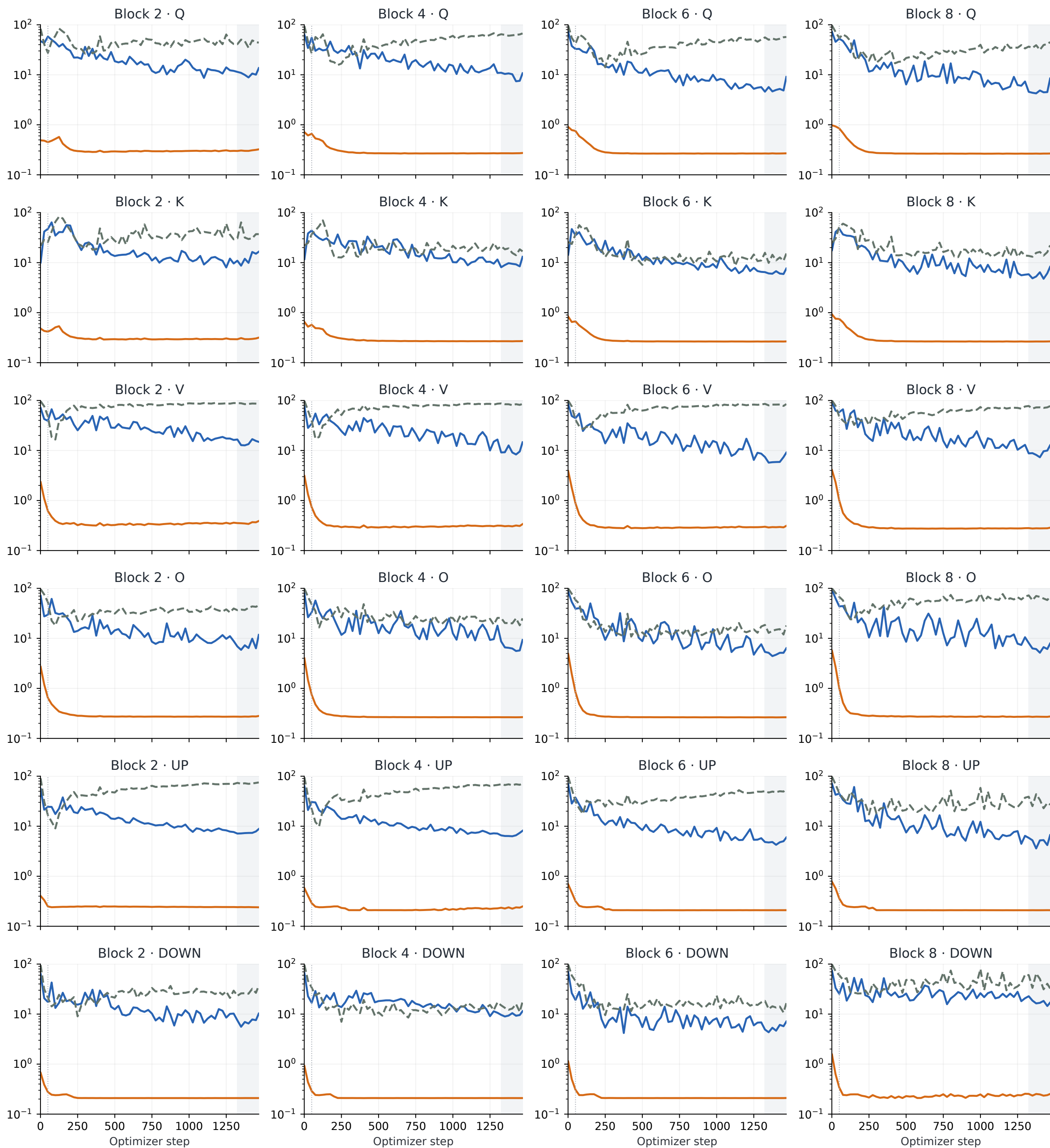
AdamW adaptive update Muon post-NS update Muon pre-NS momentum



60 snapshots per object. Each matrix normalized separately; LR/decay excluded. Shading: final-10% cooldown.

Energy in the leading singular direction · all 24 matrices

AdamW adaptive update Muon post-NS update Muon pre-NS momentum



$100 \times \sigma_1^2 / ||\text{matrix}||^2_F$. Lower values mean less concentration in one direction, not necessarily better learning.

Magnitude of the adaptive parameter step · all 24 matrices

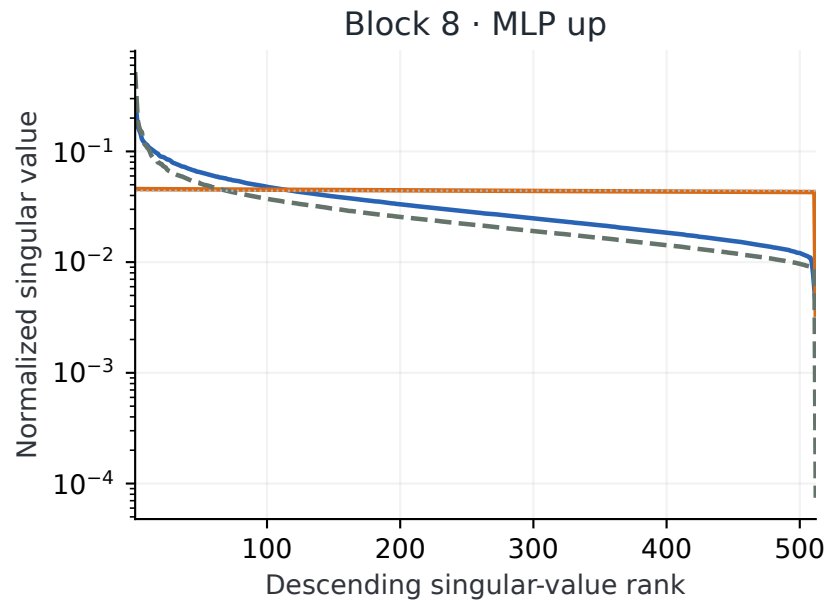
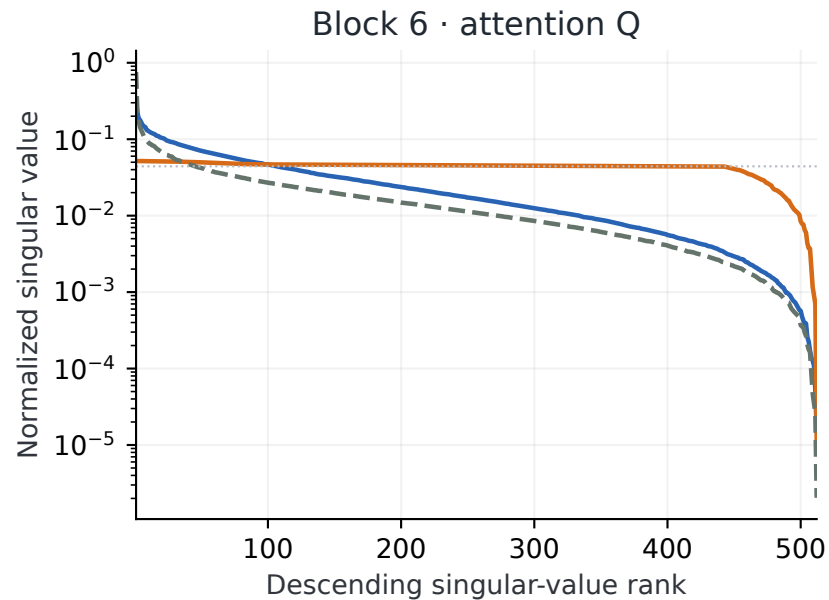
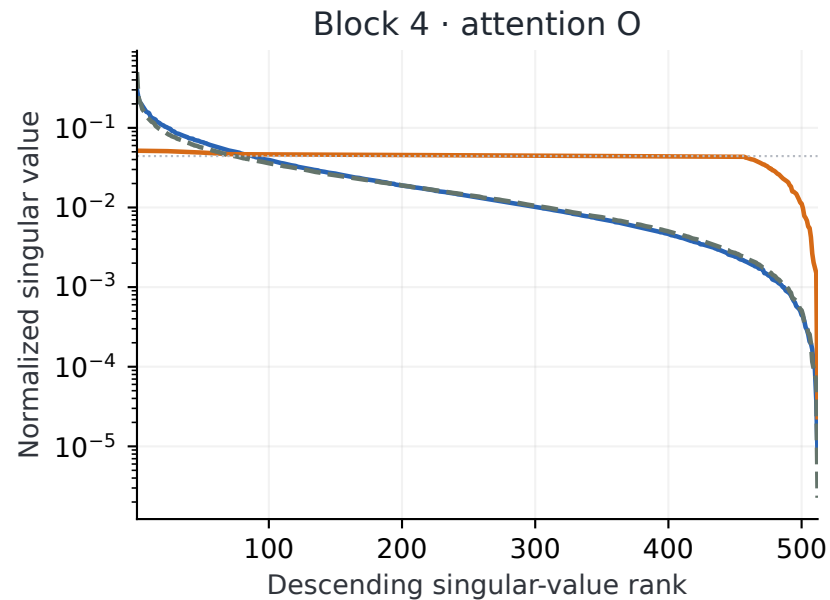
AdamW adaptive update Muon post-NS update



$||LR \times \text{adaptive direction}||_F$, before weight decay and parameter-write rounding. Group learning rates differ.

Full spectra at the final update · step 1469

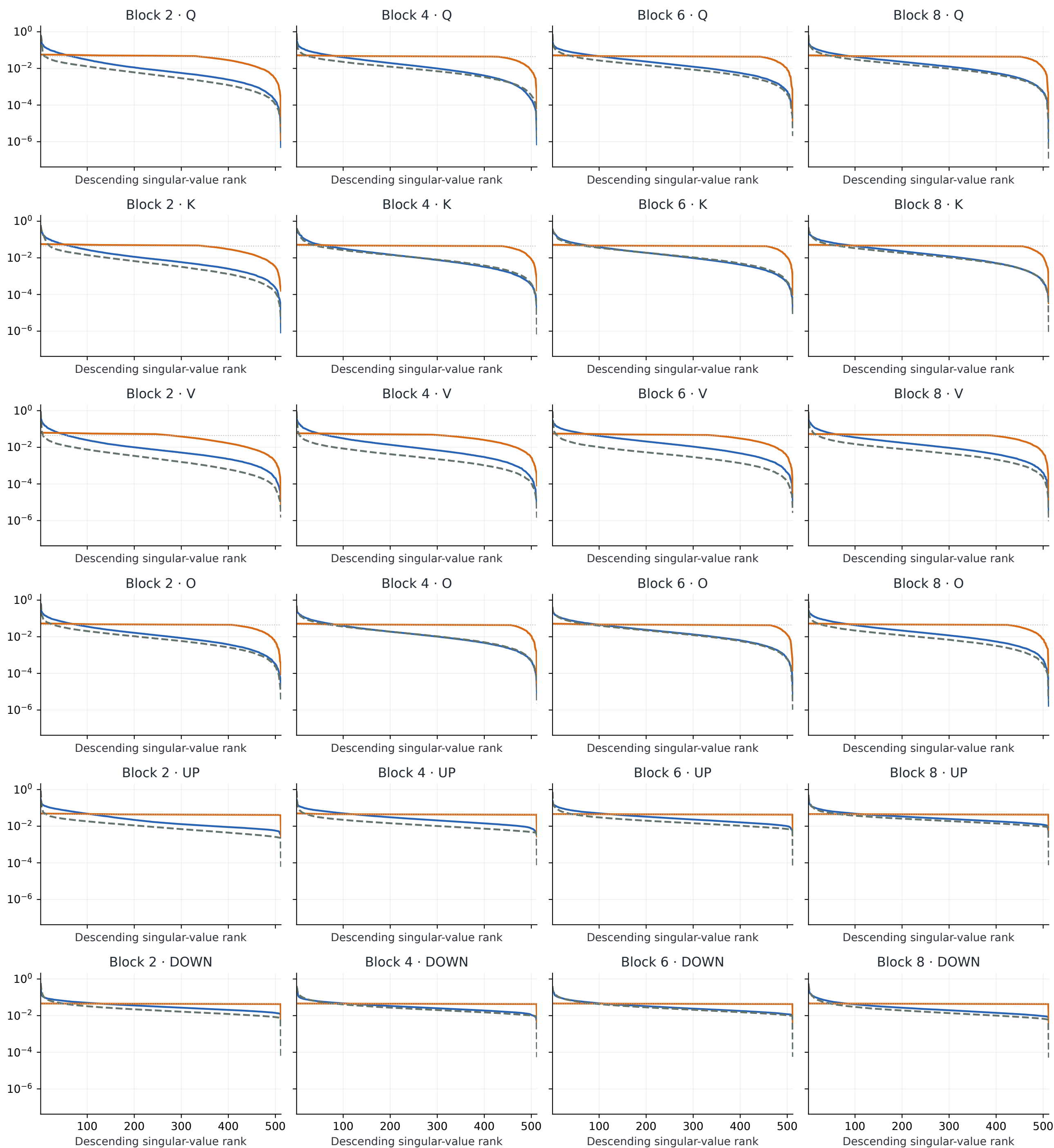
AdamW adaptive update Muon post-NS update Muon pre-NS momentum



All 512 singular values are shown. Each matrix has unit Frobenius norm; leading singular values are retained.

Full spectra at the final update · step 1469

AdamW adaptive update Muon post-NS update Muon pre-NS momentum



All 512 singular values are shown. Each matrix has unit Frobenius norm; leading singular values are retained.